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Dielectric Relaxation of Ethylene Carbonate and Propylene Carbonate from Molecular Dynamics Simulations

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Ethylene carbonate (EC) and propylene carbonate (PC) are widely used solvents in Li-ion batteries and supercapacitors. Ion dissolution and diffusion in those media are correlated with solvent dielectric responses. Here, we use all-atom molecular dynamics simulations of the pure solvents to calculate dielectric constants and relaxation times, and molecular mobilities. The computed results are compared with limited available experiments to assist more exhaustive studies of these important characteristics. The observed agreement is encouraging and provides guidance for further validation of force-field simulation models for EC and PC solvents.

Introduction

The cyclic carbonates [FIG. 1] ethylene carbonate (EC) and propylene carbonate (PC) are widely used solvents in battery applications.[1] Although EC and PC have high dielectric constants for dissociation of lithium salts, their kinetic characteristics are more sluggish than for commonly encountered aqueous solutions. Thus, in practical applications EC and PC electrolytes often incorporate other low viscosity materials to achieve faster ion transport.[1] Since batteries operate over a substantial range of temperatures, the variation in electrolyte properties with temperature is an important consideration in electrolyte development also. Experimental and simulation studies of the dielectric properties of pure EC and PC solvents at wide ranges of temperature are limited.[2–6] Here, we address the dielectric constant and dielectric relaxation times for pure EC and PC electrolytes at temperatures ranging from room temperature to 600 K, and also evaluate molecular mobilities for these solvent molecules. We carry out molecular simulation studies with all-atom force fields and compare the computed properties against available experimental results. Our goal is to assist more exhaustive experimental and modeling studies of these solvents.[7]

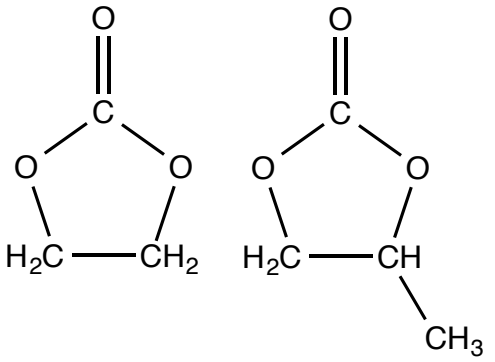


FIG. 1: Chemical structures of ethylene carbonate (EC) and propylene carbonate (PC).

Methods

Molecular dynamics simulations were carried out using the Gromacs simulation package.[8] Solvent molecules (1000 EC and 1000 PC) were placed randomly in a cubic simulation box (edge length 5 nm to 6 nm) to generate initial configurations. The OPLS-AA force field parameters were used to represent both EC and PC molecules.[9] Periodic boundary conditions were applied to mimic bulk liquid conditions. An energy minimization calculation and 1 ns of density equilibration were followed by a 50 ns production run using a constant pressure ensemble (NPT). A Nosé-Hoover thermostat[10,11] maintained the temperature, and the Parrinello-Rahman barostat[12] set the pressure at 1 atm throughout the simulation. Bonds involving hydrogen atoms were constrained using the LINCS algorithm.[13] The particle mesh Ewald method with cut-off at 1.2 nm was used to compute electrostatic interactions. Configurations were sampled every 2 ps for the dielectric constant calculations.

Analysis

Dielectric relaxation assesses the lag of the material polarization in responding to a change in an applied electric field. The delay depends on the frequency of the changing electric field. At high frequency, the dipole reorientation lags behind the field.[14] At low frequency, the polarization more nearly achieves equilibrium because the dipoles can more satisfactorily reorient to the changing field.

The frequency-dependent dielectric constant can be written as[14–16]

$$\epsilon(\omega) = \epsilon'(\omega) - i\epsilon''(\omega). \tag{1}$$

The imaginary component, the dielectric loss, describes energy lost as heat. In terms of the dipole moment autocorrelation function,

$$P(t) = \frac{\langle M(0)M(t)\rangle - \langle M(0)\rangle^2}{\langle M(0)^2\rangle - \langle M(0)\rangle^2}\ , \tag{2}$$

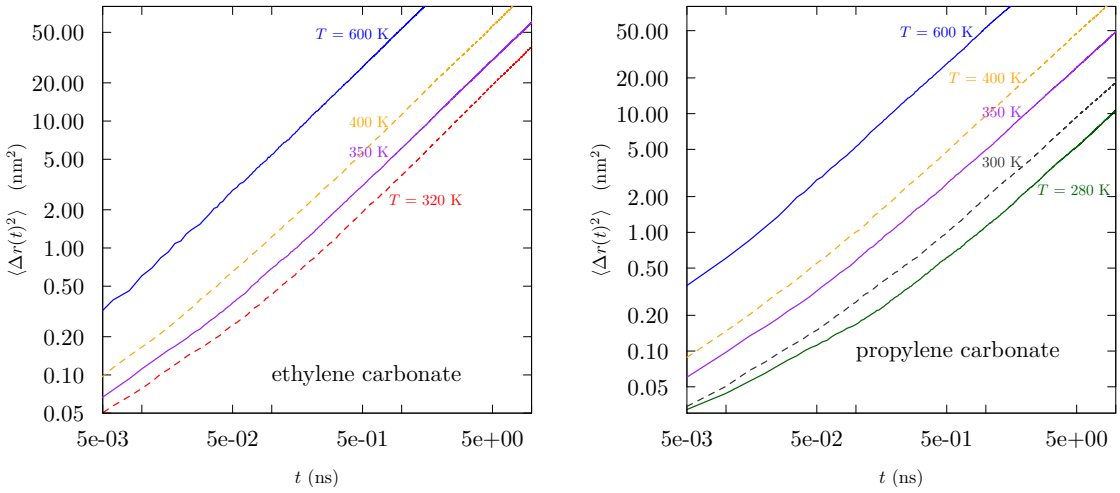


FIG. 2: Mean-squared displacements of the center of mass of PC and EC as a function of elapsed time at several temperatures.

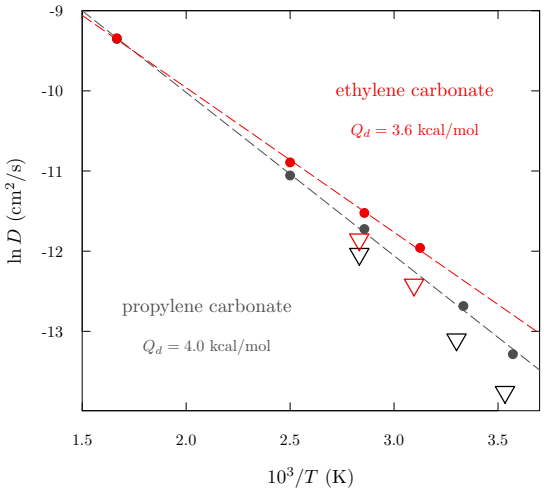


FIG. 3: Comparison of solvent molecule self-diffusion coefficients estimated from simulations with available experiment on $LiPF_6$ solutions. Q_d indicates a heat of activation for this process, $\ln D \propto -Q_d/T$, identifying the slope of the indicated fitted lines.

$\epsilon''(\omega)$ is given by the cosine transform[2,16,17]

$$\frac{\epsilon''(\omega)}{\epsilon(\omega=0)-\epsilon_0} = \omega \int_0^\infty P(t)\cos(\omega t)\,\mathrm{d}t \ . \qquad (3)$$

Here $M(t)$ is the dipole moment of the simulated system at time t, ϵ_0 is the permittivity of free space, and the relative electric permittivity $\epsilon(\omega=0)/\epsilon_0$ is the static dielectric constant. The formula Eq. (3) explicitly recognizes the absence of electronic polarizibility in the present simulation model, and thus sets the traditional infinite frequency relative permittivity to unity; $\epsilon(\omega=\infty)/\epsilon_0 = 1$.

To calculate the dipole autocorrelation function, $P(t)$ (Eq. (2)), separate 1 ns simulations were obtained, saving configurations every 1 fs to achieve satisfactory time resolution. Other simulation conditions were the same as above.

We fit the computed $P(t)$ to a Kohlrausch-Williams-Watts (KWW) model,

$$P_{\mathrm{KWW}}(t) = \exp[-(t/\tau)^\beta] \ . \qquad (4)$$

The case of $\beta = 1$ corresponds to Debye relaxation. The available experiments suggest that Debye relaxation is a helpfully accurate approximation. We take the difference of $1-\beta$ as a characterization of the importance of non-Debye relaxation. The temperature dependence of the relaxation time may be modeled as

$$\tau^{-1} = A\exp(-H^*/k_{\mathrm{B}}T) \ , \qquad (5)$$

with H^* a heat of activation taken here to be independent of T. Though specific forms of the pre-exponential factor A are debated,[2,18,19] we consider that factor a constant parameter for these results. Then H^* can be evaluated from the $1/T$ slope of the Arrhenius plot

$$\ln\tau \propto H^*/k_{\mathrm{B}}T \ . \qquad (6)$$

Results

The mobilities are obtained standardly from the mean-squared displacements of the center of mass of the solvent molecules (FIG. 2). Experimental values are apparently only available for solutions with $LiPF_6$ at substantial concentrations. Still, computed results for these force fields

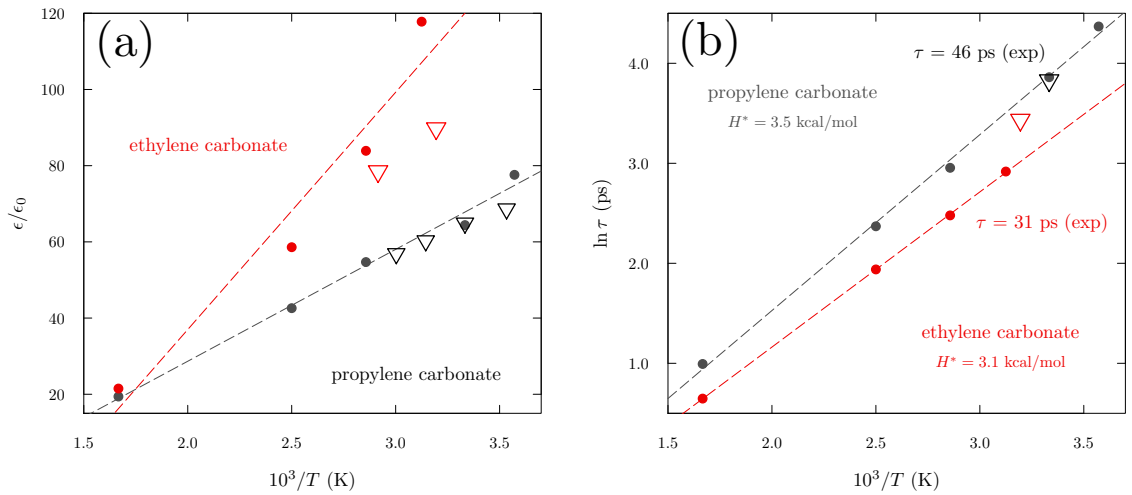


FIG. 4: (a) Static dielectric constants ϵ/ϵ_0, and (b) relaxation times τ for EC and PC at several temperatures. Experimental data is shown in open triangles.[2] The indicated experimental relaxation times are the ones provided in the figure captions of that experimental report.[2]

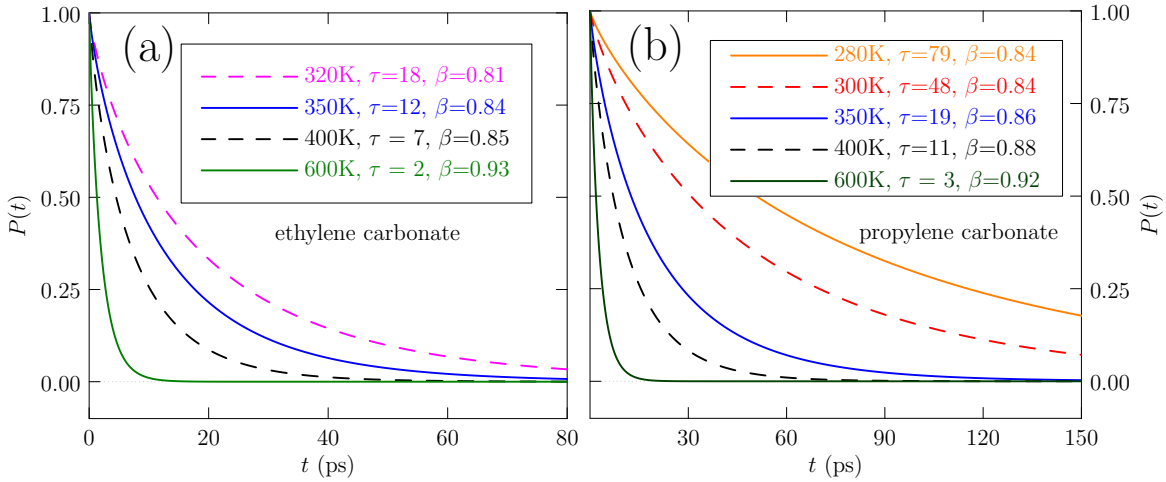


FIG. 5: Dipole auto-correlation functions for EC (left) and PC (right) at several temperatures. The indicated relaxation times are in ps; see TABLE I.

disagree with experiment by less than 50% (FIG. 3), and the comparison is thus encouraging.

The agreement of computed static dielectric constants ϵ/ϵ_0 (FIG.4) with experimental values is good for PC; for EC at the lowest temperatures, the discrepancy from experiment is about 30%. The relaxation times (FIG.4) agree reasonably with experiment, though we note that the experimental values were extracted from experiment assuming Kirkwood and Debye models for the dielectric characteristics of these solvents. The values of the relaxation times, *e.g.*, 46 ps for PC and 31 ps for EC, compared to 7.5 ps for liquid water at T=303K,[20] highlight the fact that these solvents are comparatively sluggish. Together with the mobility results, these times emphasize that simulations of several $\times$10 ns are required to obtain sound results for these kinetic properties. Numerical results are collected in TABLE I.

Dipole auto-correlation functions calculated for the simulation trajectory (FIG. 5) are fitted to Eq. (4) with stretch parameter β. $1-\beta$ is typically of rough magnitude 0.1 (TABLE I) and not highly sensitive to temperature, but trending to smaller values with higher temperatures. This suggests that a Debye model of relaxation is helpfully accurate here and indeed that has been the outlook from experiments.[2] The dielectric relaxation times are remarkably long on conventional simulation times even though these are small-molecule liquids. Still the agreement with experiment [FIG. 4(b)] is encouraging: τ =

	T (K)	ρ (kg/dm^3)	ϵ/ϵ_0	τ (ps)	β
EC	320	1.305	118±7	17	0.80
	350	1.275	84±5	12	0.84
	400	1.223	59±6	7	0.85
	600	0.997	22±1	2	0.93
PC	280	1.246	78±5	79	0.84
	300	1.226	64±4	48	0.84
	350	1.175	55±3	19	0.86
	400	1.124	43±1	11	0.88
	600	0.894	19±1	3	0.92

TABLE I: Static dielectric constant ϵ/ϵ_0, relaxation time τ, and stretch parameter β of EC and PC as a function of temperature. See Eq. (4). The molecular models studied here are flexible, but do not include electronic polarizability. Because electronic polarizability is not included, mean dipole moments are larger than the isolated molecule dipole moments; 5.47 D (for EC at $T = 320$K), and 6.47 D (for PC at $T = 300$K). The observed mean dipole moments are insensitive to T. The fitted stretch parameter yields $1-\beta$ typically about 0.1, not highly sensitive to T, but trending to smaller values with higher temperatures.

46 ps from experiment on PC at room temperature compared to $\tau = 48$ ps from simulations. The extracted activation parameters H^* are 3.1 kcal/mol and 3.5 kcal/mol for EC and PC [FIG. 4(b)]; an experimental value is 2.1 kcal/mol for PC.[2] Note that extraction of relaxation times from experiment utilized values for factors assessing electronic polarizability $\epsilon(\omega = \infty)/\epsilon_0$, which we deliberately avoid here.

Analysis of experimental dielectric relaxation times was improved by fitting to a Vogel-Fulcher-Tamman model,[2] $\propto \exp\{-B/(T - T_0)\}$ with fitting parameters B and T_0, and also for the non-zero-concentration mobilities that we utilized above (FIG. 3).[7] Then the dielectric relaxation times nicely tracked the experimental viscosity.[2] But the fitting of the experimental dielectric relaxation results for PC were probably sensitive to one super-cooled ($T = -78$C) case. The present computational results cover a higher and broader temperature range and complications beyond a single heat of activation are not required. Still it would be interesting, and maybe of practical relevance, for subsequent experiment and modeling to investigate super-cooled conditions of EC/PC more thoroughly.

Conclusion

The OPLS-AA force field parameters for EC and PC provide reasonable agreement with the available experimental data on pure solvent dielectric properties, both for the static dielectric constant and the dielectric relaxation times. This study provides a benchmark for force field parameters for pure EC and PC solutions and can be extended to study the electrical properties of mixed electrolyte solutions.

Acknowledgements

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